

HW - Section 2.4 Write-up

31. If G is a group with operation $\circ$ and $a, b, c \in G$, then $a \circ b = c \circ b$ implies $a = c$.

Proof.

Let $a \circ b = c \circ b$. Multiply both sides by b^{-1}
 $(a \circ b) \circ b^{-1} = (c \circ b) \circ b^{-1}$. By Associative law we get
 $a \circ (b \circ b^{-1}) = c \circ (b \circ b^{-1})$. So $a \circ e = c \circ e$. Which proves $a = c$. $\square$

38. What property identifies the zero divisors in $\mathbb{Z}_6$?
 $x \neq 0$ is a zero divisor in $\mathbb{Z}_6$ if and only if x and 6 have a common divisor greater than 1.

Thus zero divisors in $\mathbb{Z}_6$ are 2, 3 and 4.

It appears that the nonzero elements of $\mathbb{Z}_6$ are partitioned into the set of zero divisors and the set $U(6)$.

44. Prove that if $\mathbb{Z}_n$ has zero divisors, then right cancellation does not hold for all elements of $\mathbb{Z}_n$.

Proof

Let a and b be zero divisors in $\mathbb{Z}_n$. This means $a \neq 0$ and $b \neq 0$ but $a \cdot b = 0$. We also know $0 \cdot b = 0$. So we have $a \cdot b = 0 \cdot b$ but $a \neq 0$ so right cancellation fails.

extra List all idempotents and zero divisors in $\mathbb{Z}_{10}$

Idempotents are 0, 1, 5, and 6

Zero divisors are 2, 4, 5, 6, and 8